Output:-

Que :- Consider a disk queue with requests for l/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The SSTF scheduling algorithm is used. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199. Write a Program to calculate the total head movement (in number of cylinders) incurred while servicing these requests.

Code :-

```
#include <stdio.h>
#include <stdlib.h>

int main() {
   int req[] = {98,183,41,122,14,124,65,67};
   int n = 8;
   int head = 53;
   int total = 0;
   int visited[8] = {0};

   for(int i = 0; i < n; i++) {
      int min = 9999, index = -1;

      for(int j = 0; j < n; j++) {
         if(!visited[j]) {
            int dist = abs(head - req[j]);
            if(dist < min) {
               min = dist;
               index = j;
            }
         }
      }

      visited[index] = 1;
      total += min;
      head = req[index];
   }

   printf("Total Head Movement = %d\n", total);

   return 0;
}
```

Output:-

```
hari_ai@Harryfunlapcare: ~
hari_ai@Harryfunlapcare:~$ nano sstf.c
hari_ai@Harryfunlapcare:~$ gcc sstf.c -o sstf
hari_ai@Harryfunlapcare:~$ ./sstf
Total Head Movement = 323
hari_ai@Harryfunlapcare:~$
```